# Efficacy of Antioxidants as Treatment in a Parkinson Disease Model in C. *elegans*

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## Background

Parkinson's disease (PD) is a progressive neurodegenerative condition characterized by motor dysfunction, inflammation, and reduced dopamine levels in the brain. Pathologically, PD involves the misfolding and aggregation of presynaptic proteins called alpha-synucleins (α -synucleins), which normally help maintain neuronal function. These misfolded proteins cluster into structures called Lewy bodies, which disrupt cellular processes and contribute to neuronal death.[1] While genetics is one cause of PD, exposure to the neurotoxin rotenone, commonly used to model PD in laboratory models, has been shown to trigger α-synuclein misfolding and induce PD-like symptoms.[3]

Recent studies have explored the therapeutic potential of nutrition in neurodegenerative diseases. Diets rich in antioxidants, particularly the Mediterranean diet, have demonstrated neuroprotective effects. Evidence suggests that antioxidants can bind to α-synuclein and prevent its aggregation into Lewy bodies, potentially offering preventative or therapeutic benefits for PD.[4]

This study aimed to investigate whether the antioxidants in extra virgin olive oil (EVOO) could reduce PD-like symptoms in a *Caenorhabditis elegans* (C. *elegans*) model. Specifically, it was tested whether EVOO was more effective in preventing or alleviating symptems after onset.[5] Since antioxidants are hypothesized to bind α-synuclein and prevent Lewy body formation, it was expected that EVOO would be more effective as a preventative treatment.

To model PD, OW13 C. *elegans*, a transgenic strain engineered to express human α -synuclein, combined with rotenone to induce PD-like pathology. Given that PD is associated with shortened lifespan, impaired mobility, and increased α-synuclein aggregation, the study used three assays, a lifespan assay, thrashing assay, and fluorescence microscopy, to evaluate the potential benefits of EVOO.[3]

Five experimental groups of C. *elegans* were established: a control group, a rotenone-only group, an EVOO-only group, a pre-treatment group (exposed to EVOO before rotenone), and a co-treatment group (exposed to EVOO and rotenone simultaneously). All worms were fed *Escherichia coli* (E. *coli*) OP50 as a standard food source. These conditions were designed to evaluate the protective and therapeutic effects of EVOO in a PD model.[2]

## Results

### ***A Moderate Concentration of Rotenone (0.5 mM) Induces PD-Like Symptoms Without Immediate Lethality***

Rotenone is a neurotoxin known to induce PD symptoms by triggering α-synuclein misfolding and aggregation, leading to neuronal dysfunction and death.[3] Although its toxicity is well documented, the precise concentration required to induce PD-like symptoms in C. *elegans* remains unclear. In other research studies that utilized rotenone to cause α-synuclein aggregation, it was noticed that various rotenone concentrations were ingested by C. *elegans*, causing different severities in effects.[3]

To ensure consistency in the upcoming three experiments, it was crucial to find a specific concentration that was enough to decrease C. *elegans* lifespan and mobility, but not too high to kill off entire populations immediately. Although OW13 C. *elegans* were intended for use, they were unavailable during preliminary testing. However, wild-type C. *elegans* were used as a substitute due to their nearly identical lifespan and mobility characteristics.

The three experimental conditions studied were 0.25 mM, 0.5 mM, and 0.75 mM of rotenone, which was diluted in dimethyl sulfoxide (DMSO). Wild-type C. *elegans* at the L4-stage were placed on nematode growth medium (NGM)/5-fluorodeoxyuridine (FUDR) plates seeded with *E. coli* OP50 mixed with the rotenone solution and incubated at 25°C.

Using a dissecting microscope, the C. *elegans* in the plates were observed and photographed, and their movements were captured on video 1 day and 3 days after the rotenone exposure. In all plates, C. *elegans* showed slower movement, with periods of reversals (moving backwards). The 0.25 mM rotenone group exhibited the least mortality and qualitatively retained the fastest movement. In contrast, the 0.75 mM rotenone group had the highest mortality and showed minimal movement. Worms exposed to 0.5 mM rotenone displayed slower, more hesitant movements than the 0.25 mM group but fewer deaths than the 0.75 mM group.

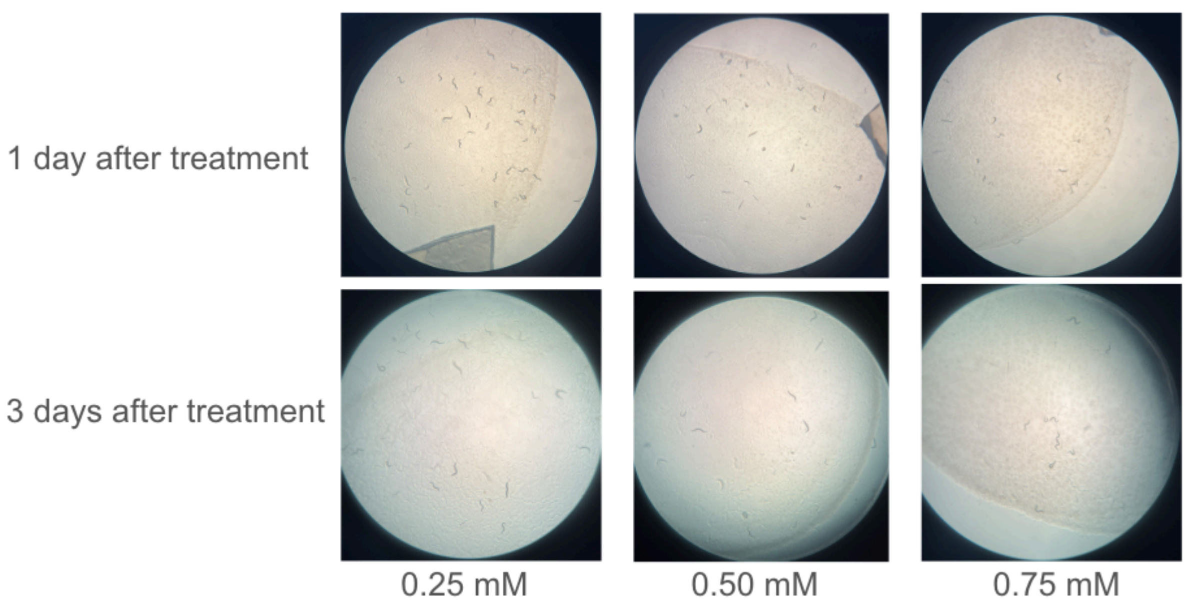

**Figure 1. Wild Type *C. elegans* preliminary cultures after rotenone treatment.** Images from the dissecting microscope (10x resolution) are representative of each sample population. L4 C. *elegans* were transferred to NGM/FUDR plates seeded with E. *coli* OP50 and a various rotenone and DMSO solution and incubated at 25°C. The change in the total number of the worms in each plate were qualitatively observed by photographing the plates through a dissecting microscope 1 day and 3 days after being exposed to rotenone.

***Issues with Synchronization Led to the Termination of the Lifespan Assay of the PD Model***

To determine the impact of EVOO and rotenone on the longevity of OW13 C. elegans, a lifespan assay was conducted. It was hypothesized that EVOO would protect against rotenone-induced lifespan reduction by preventing α-synuclein aggregation. To test this, 15 NGM/FUDR plates were divided into five treatment groups: control, rotenone-only, EVOO-only, pre-treatment, and co-treatment.

OW13 C. elegans were synchronized using M9 bleaching to ensure uniform developmental stages. Approximately 0.5 mL of egg suspensions were added to each plate, and worms were incubated at 25°C for two days. However, there were challenges behind the synchronization process of the OW13 strain. On multiple occasions, no worms or eggs were observed on the experiment plates. Possible errors could include worms not detaching from the plate surface during M9 buffer washing or a loss of eggs during pipetting or accidental discard with biohazardous waste. Without a synchronized population of OW13 nematodes, the lifespan assay was unable to proceed. Observations under the dissecting microscope confirmed the absence of viable eggs or worms, preventing the collection of meaningful data for the lifespan assay.

***EVOO Pre-Treatment Partially Restores Locomotion in OW13 C. elegans Exposed to Rotenone***

To evaluate the impact of EVOO and rotenone on the locomotion of OW13 C. *elegans*, the thrashing assay was performed. Locomotor function is a critical measure of PD-like symptoms, and this assay aimed to quantify these effects across different experimental conditions. The study explored whether EVOO could alleviate rotenone-induced motor impairments.

The same five treatment groups, control, rotenone-only, EVOO-only, pre-treatment, and co-treatment, were created using NGM/FUDR plates given their specific concentrations of EVOO and rotenone. Similarly to the lifespan assy, the thrashing assay was prepared by synchronizing the OW13 C. *elegans*.

Due to challenges with nematode synchronization, worms were chunked directly from an existing OW13 culture onto the experimental plates. While the C. *elegans* were exposed to the treatment for two days before analysis, the pre-treatment group had one-day exposure to EVOO and one-day exposure with the addition of rotenone. Following exposure and incubation at 25°C, nematodes were transferred to wells containing M9 buffer for thrashing activity measurements. Using a dissecting microscope, thrashing movements of 9-10 worms per condition were recorded for 30 seconds each.

The control and EVOO-only groups displayed the highest thrashing rates ($27.2 \pm 6.5$ and $24.4 \pm 7.2$ respectively), while the rotenone-only treatment group exhibited slower, less frequent movements ($8.6 \pm 4.4$). Pre-treatment with EVOO showed partial restoration of locomotion ($18.8 \pm 6.5$), outperforming the co-treatment group, which displayed the slowest movement (3.9

± 3.3). These results suggest that EVOO is more effective as a preventative than a therapeutic measure.

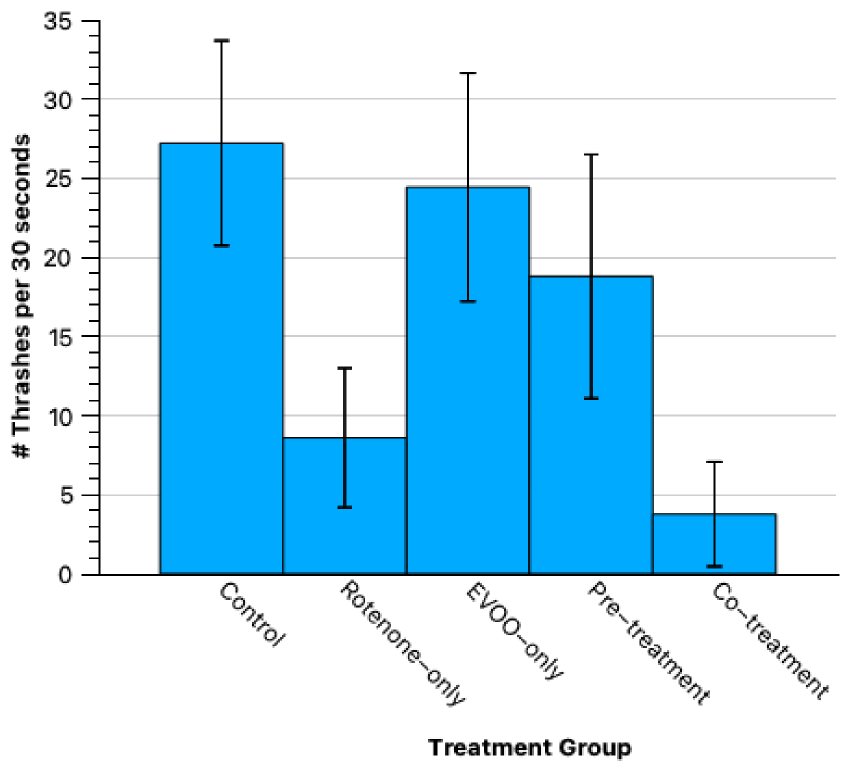


**Figure 2. Average thrashing rate of OW13 C. *elegans* across treatment groups.** Thrashes. Chunked OW13 C. *elegans* were transferred to NGM/FUDR plates seeded with E. *coli* OP50 and specific concentrations of EVOO and rotenone. After 2 days of incubation at 25°C, 9–10 worms per group were transferred into the M9 buffer, and thrashing movements were counted over 30-second intervals using a dissecting microscope. Bars represent the mean number of thrashes per 30 seconds while error bars indicate standard deviation.

***Fluorescence Imaging Highlights Reduced α-Synuclein Aggregation with EVOO Pre-Treatment***

To evaluate the impact of EVOO and rotenone on the aggregation of α-synuclein in OW13 C. *elegans*, the fluorescence assay was conducted. α-synuclein aggregation is a critical measure of PD-like pathology, and this assay aimed to visualize these YFP-tagged presynaptic proteins under different experimental conditions. The study explored whether EVOO could mitigate rotenone-induced aggregation.

Similarly to the thrashing assay, the same five treatment groups were prepared, and OW13 C. *elegans* were chunked onto the experimental plates.

To visualize fluorescence, an EVOS fluorescence imaging system was used. Imaging was performed with GFP lighting in a dark box, set to 60% light intensity to enhance the visibility of YFP-tagged aggregation without overexposing the images.

Using the EVOS system, fluorescence patterns of approximately 3 worms per condition were captured and analyzed. Worms exposed to rotenone alone exhibited larger, brighter fluorescent aggregates compared to those on control plates, reflecting increased alpha-synuclein aggregation due to rotenone-induced neurotoxicity. Plates containing EVOO showed fewer and less intense fluorescent aggregates, suggesting potential protective effects. The pre-treatment group exhibited reduced aggregation compared to the co-treatment group, consistent with the hypothesis that EVOO may be more effective in preventing aggregation than reversing it.

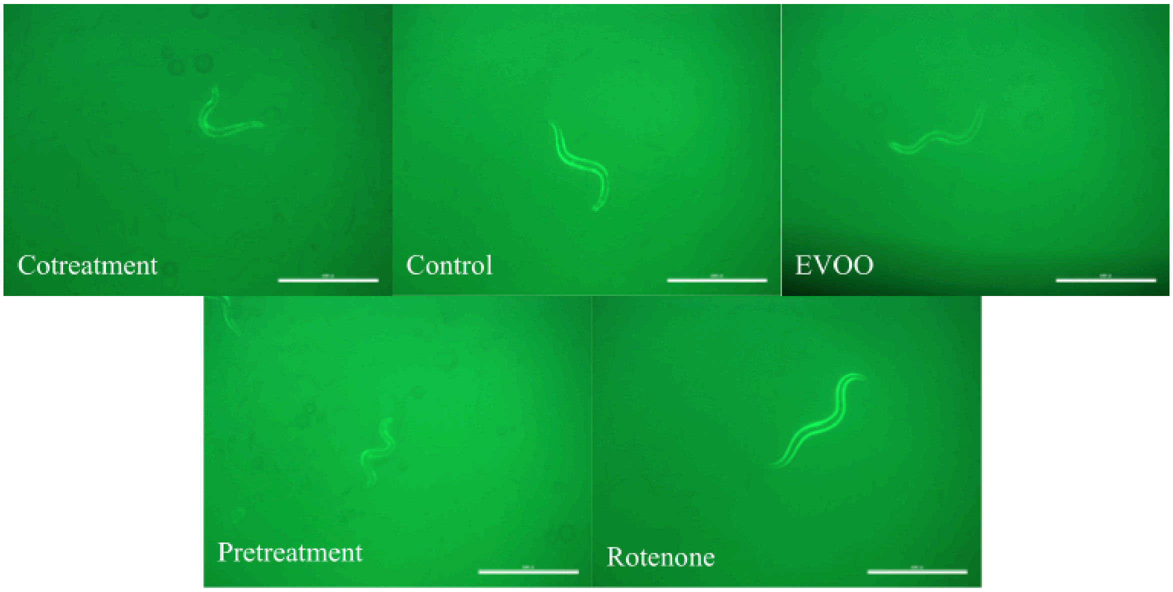


**Figure 3. Fluorescence microscopy of *C. elegans* expressing α-synuclein::YFP.** Images were captured directly from the agar plates using the EVOS M5000 microscope at 40x total magnification. Worms were photographed directly on the plates to avoid disturbing their natural position. Although the worms expressed YFP, the GFP protein setting was used for imaging; the YFP signal was still clearly visible under these conditions. A consistent 60% light intensity was applied across all samples, and representative images were selected to reflect the general appearance of each treatment group (scale bar = 1000 µm).

**Discussion**

The project aimed to investigate whether antioxidants in EVOO could prevent or alleviate PD-like symptoms in C. *elegans*. It was hypothesized that EVOO would be more effective at preventing PD than reversing them, based on the idea that antioxidants may bind to α-synuclein and prevent its aggregation into Lewy bodies.

Preliminary testing established that 0.5 mM rotenone effectively induced PD-like symptoms in wild-type C. *elegans* without causing excessive mortality. This concentration resulted in moderate motor impairment and survivability, providing a reliable baseline for the upcoming experiments using the OW13 transgenic strain.

The results suggest that EVOO pre-treatment can reduce the severity of PD-related symptoms. In the thrashing assay, worms pre-treated with EVOO demonstrated significantly greater locomotion than those exposed to rotenone alone or co-treated, indicating improved neuromuscular function. Fluorescence imaging also revealed reduced α-synuclein aggregation in the pre-treated group, supporting the hypothesis that EVOO interferes with early misfolding and aggregation processes.

The co-treatment with EVOO showed moderate benefits but was less effective than pre-treatment, reinforcing the idea that EVOO works best as a preventative measure rather than as a therapeutic intervention. The lifespan assay yielded inconclusive results, which limited the ability to assess long-term protective effects. However, the general trends observed across behavioral and fluorescence assays provide preliminary support for EVOO's neuroprotective role in this model.

Due to the unavailability of OW13 worms during preliminary testing, wild-type C. *elegans* were used as substitutes. Although they share many physiological characteristics, their response to rotenone may not fully reflect that of the mutant strain. Furthermore, age variability caused by synchronization issues and the need to chunk worms mid-experiment introduced inconsistency in developmental stages across samples. Since PD-like symptoms intensify with age in C. *elegans*, this variability may have influenced behavioral outcomes and aggregation levels. Additionally, mobility assessments relied on qualitative observations under a microscope rather than quantitative tracking methods, which may have limited precision in measuring neuromuscular function.

Future research should aim to replicate the experiment using both wild-type and OW13 worms across different rotenone concentrations to better understand how genetic differences affect susceptibility to neurotoxicity. Synchronization protocols should be refined to ensure consistent age distribution and more reliable data collection, particularly for lifespan analysis. Quantitative image analysis tools such as ImageJ could be used to measure α-synuclein aggregation intensity across treatment groups. Further studies could also test different concentrations of EVOO to determine the most effective dose and compare its neuroprotective potential to other antioxidant compounds such as curcumin or resveratrol.

This study demonstrates the potential of EVOO as a preventative strategy against PD. By reducing motor impairments and α-synuclein aggregation, it highlights the role of dietary antioxidants in promoting brain health and shaping future therapeutic approaches.

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